Chapter 3 Additional topics in probability

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3.1 Introduction

- Some special distributions that have useful applications in statistics
- Joint distributions of random variables
- Distributions of functions of random variables
- Two limit theorems
  - The law of large numbers
  - The central limit theorem

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Bernoulli probability distribution: The probability function:

$$p(x) = P(X = x) = \begin{cases} p^x(1-p)^{(1-x)}, \; x = 0, 1 \\ \qquad\qquad\qquad 0, \text{ otherwise} \end{cases}$$

- Only two possible outcomes, *e*.g., coin tossed,
  - "success" if heads occurs, taking value 1 with probability p
  - "failure" if tails occurs, taking value 0 with probability $q = 1 - p$
- $E(X) = p$
- $Var(X) = pq$
- $M_X(t) = pe^t + (1 - p)$

3.2.1 The binomial probability distribution

Definition 3.2.1 A binomial experiment is one that has the following properties:

- The experiment consists of n identical trials
- Each trial results in one of the two outcomes, called a success S and failure F
- The probability of success on a single trial is equal to p and remains the same from trial to trial. The probability of failure is $q = 1 - p$.
- The outcomes of the trials are independent
- The random variable X is the number of successes in n trials.

The number of ways of obtaining x successes in n trials is given by:

$$\binom{n}{x} = \frac{n!}{x!\,(n-x)!}$$

3.2.1 The binomial probability distribution

Binomial probability distribution with parameters (n, p): In a succession of Bernoulli trials, the probability of observing exactly x successes in n independent Bernoulli trials

- pmf

$$P(X = x) = p(x) = \begin{cases} \begin{pmatrix} n \\ x \end{pmatrix} p^x q^{(n-x)}, \ x = 0, 1, \dots, n, 0 \le p \le 1 \\ 0, \text{ otherwise} \end{cases}$$

  - Denoted by $b(x; n, p)$

- Cumulative probability distribution

$$B(x; n, p) = \sum_{i=0}^{x} b(i; n, p)$$

3.2.1 The binomial probability distribution

- Mean

$$E(X) = \mu = np$$

- Variance

$$Var(X) = \sigma^2 = np(1-p)$$

- Moment-Generating Function

$$M_X(t) = [pe^t + (1-p)]^n$$

3.2.1 The binomial probability distribution

Example 3.2.1 It is known that screws produced by a certain machine will be defective with probability 0.01 independent of one another. If we randomly pick 10 screws produced by this machine, what is the probability that at least two screws will be defective?

3.2.1 The binomial probability distribution

Example 3.2.1 It is known that screws produced by a certain machine will be defective with probability 0.01 independent of one another. If we randomly pick 10 screws produced by this machine, what is the probability that at least two screws will be defective?

Solution Let X be the number of defective screws out of 10. Then X can be considered as a binomial random variable with parameters $(10, 0.01)$. Using Definition 3.2.2, we obtain that at least two screws will be defective as:

$$P(X \geq 2) = \sum_{x=2}^{10} \binom{10}{x} (0.01)^x (0.99)^{10-x}$$

$$= 1 - [P(X=0) + P(X=1)] = 0.004$$

- R-command: 1-pbinom(1,10,0.01)

Example 3.2.2 Suppose we know that the frequency of a dominant gene, A, in a population is 0.2. If we randomly select 8 members of this population, what is the probability that at least 6 of them will display the dominant phenotype? Assume that the population is sufficiently large that removing eight individuals will not affect the frequency and that the population is in Hardy-Weinberg equilibrium.

Example 3.2.2 Suppose we know that the frequency of a dominant gene, A, in a population is 0.2. If we randomly select 8 members of this population, what is the probability that at least 6 of them will display the dominant phenotype? Assume that the population is sufficiently large that removing eight individuals will not affect the frequency and that the population is in Hardy-Weinberg equilibrium.

Solution

First of all, note that an individual can have the dominant gene, A, if the person has traits AA, aA, or Aa. Hence, if the gene frequency is 0.2, the probability that an individual is of genotype A is:

$$P(A) = P(AA \cup Aa \cup aA) = P(AA) + 2P(Aa)$$

$$= (0.2)^2 + 2(0.2)(0.8) = 0.36$$

(Solution continued)

Let X denote the number of individuals out of 8 that display the dominant phenotype. Then X is binomial with $n = 8$, and $p = 0.36$. Thus, the probability that at least 6 of them will display the dominant phenotype is:

$$\begin{aligned} P(X \geq 6) &= P(X = 6) + P(X = 7) + P(X = 8) \\ &= \sum_{i=6}^{8} \binom{8}{i} (0.36)^i (0.64)^{8-i} \\ &= 0.029259 \end{aligned}$$

R-command: $1 - \text{pbinom}(5, 8, 0.36)$

3.2.1 The binomial probability distribution

Example 3.2.3 A manufacturer of inkjet printers claims that only 5% of their printers require repairs within the first year. If, of a random sample of 18 of the printers, 4 required repairs within the first year, does this tend to refute or support the manufacturer's claim?

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Solution Let us assume that the manufacturer's claim is correct; that is, the probability that a printer will require repairs within the first year is 0.05. Suppose 18 printers are chosen at random.

3.2.1 The binomial probability distribution

(Solution continued)

Let X represent the number of printers that require repair within the first year. Then X follows the binomial probability mass function (pmf) with $p = 0.05$ and $n = 18$. The probability that four or more of the 18 will require repair within the first year is given by:

$$P(X \geq 4) = \sum_{x=4}^{18} \binom{18}{x} (0.05)^x (0.95)^{18-x}$$

or, using the binomial table:

$$P(X \geq 4) = 1 - B(3; 18; 0.05) = 1 - 0.9891 = 0.0109$$

This value (approximately 1.1%) is very small. We have shown that if the manufacturer's claim is correct, then the chances of observing four or more bad printers out of 18 are very small. But we did observe exactly four bad ones. Therefore, we must conclude that the manufacturer's claim cannot be substantiated.

3.2.2 Poisson probability distribution

Poisson probability distribution with parameter $\lambda > 0$ denoted by Poisson(λ) with probability distribution:

$$P(X = x) = f(x; \lambda) = f(x) = \frac{e^{-\lambda}\lambda^{x}}{x!}, x = 0, 1, 2, ...$$

- Mean

$$E(X) = \lambda$$

- Variance

$$Var(X) = \lambda$$

- Moment-Generating function

$$M_X(t) = e^{\lambda(e^t - 1)}$$

Example 3.2.4

Let X be a Poisson random variable with $\lambda = 1/2$. Find:

- $P(X = 0)$
- $P(X \geq 3)$

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Solution:

$$P(X = 0) = p(0) = \frac{e^{1/2}(1/2)^0}{0!} = e^{1/2} = 0.60653$$

$$P(X \geq 3) = 1 - P(X < 3) = 1 - [p(0) + p(1) + p(2)]$$

$$= 1 - \left[e^{1/2} + e^{1/2}\left(\frac{1}{2}\right) + \frac{e^{1/2}\left(\frac{1}{2}\right)^2}{2!}\right] = 1 - 0.98561 = 0.01439$$

R-command: ppois(2, lambda=1/2, lower=FALSE)

3.2.2 Poisson probability distribution

The relation between poisson and binomial distributions

Theorem 3.2.3 if X is a binomial random variable with parameters n and p, then for each value $x = 0, 1, 2, \ldots$ and as $p \to 0, n \to \infty$ with $np = \lambda$ constant,

$$\lim_{n\to\infty} \binom{n}{x} p^x (1-p)^{(n-x)} = \frac{e^{-\lambda}\lambda^x}{x!}$$

- When n is large, we can use poisson distribution to replace binomial distribution

Example 3.2.5 If the probability that an individual suffers an adverse reaction from a particular medication is known to be0.001, determine the probability that, of 2000 individuals,

- exactly three
- more than two individuals will suffer an adverse reaction.

Solution

Let Y be the number of individuals who suffer an adverse reaction. Then Y is binomial with $n = 2000$ and $p = 0.001$. Because n is large and p is small, we can use the Poisson approximation with $\lambda = np = 2$.

- The probability that exactly three individuals will suffer an adverse reaction is:

$$P(Y = 3) = \frac{2^3 e^{-2}}{3!} = 0.18$$

  That is, there is approximately an18% chance that exactly three individuals out of2000 will suffer an adverse reaction.

3.2.2 Poisson probability distribution

Solution Continued

- The probability that more than two individuals will suffer an adverse reaction is:

$$P(Y > 2) = 1 - P(Y = 0) - P(Y = 1) - P(Y = 2) = 1 - 5e^{-2} = 0.323$$

Similarly, there is approximately a 32.3% chance that more than two individuals will have an adverse reaction.

R-command: `ppois(2, lambda=2, lower=FALSE)`

3.2.3 Uniform probability distribution

A random variable X is said to have a uniform probability distribution on (a, b), denoted by $U(a, b)$, the density function of X is given by:

$$f(x) = \begin{cases} \dfrac{1}{b-a}, \ a \leq x \leq b \\ 0, \ \text{otherwise} \end{cases}$$

The cumulative distribution function (cdf) is given by:

$$F(x) = \int_{-\infty}^{x} f(t)dt = \begin{cases} 0, \ x < a \\ \dfrac{x-a}{b-a}, \ a \leq x < b \\ 1, \ x \geq b \end{cases}$$

3.2.3 Uniform probability distribution

Example 3.2.6 If X is a uniformly distributed random variable over the interval $(0, 10)$, calculate the probability that:

- $X < 3$
- $X > 6$
- $3 < X < 8$

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- $X < 3$
- $X > 6$
- $3 < X < 8$

Solution

- $P(X < 3) = \int_0^3 \frac{1}{10}\, dx = \frac{3}{10}$
- $P(X > 6) = \int_6^{10} \frac{1}{10}\, dx = \frac{4}{10} = \frac{2}{5}$
- $P(3 < X < 8) = \int_3^8 \frac{1}{10}\, dx = \frac{1}{2}$

3.2.3 Uniform probability distribution

- Mean

$$E(X) = \frac{a+b}{2}$$

- Variance

$$Var(X) = \frac{(b-a)^2}{12}$$

- Moment-Generating function

$$M_X(t) = \begin{cases} \dfrac{e^{tb} - e^{ta}}{t(b-a)}, \, t \neq 0 \\ 1, \, t = 0 \end{cases}$$

3.2.3 Uniform probability distribution

Example 3.2.7 The melting point, X, of a certain solid may be assumed to be a continuous random variable that is uniformly distributed between the temperatures 100 and $120°C$. Find the probability that such a solid will melt between 112 and $115°C$.

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Solution

The probability density function (pdf) is given by:

$$f(x) = \begin{cases} \dfrac{1}{20}, & \text{if } 100 \leq x \leq 120 \\ 0, & \text{otherwise} \end{cases}$$

Hence,

$$P(112 \leq X \leq 115) = \int_{112}^{115} \frac{1}{20}\, dx = \frac{3}{20} = 0.15$$

Thus, there is a 15% chance of this solid melting between 112 and $115°C$.

3.2.4 Normal probability distribution

- The normal probability distribution was introduced by the French mathematician Abraham de Moivre in 1733
- Later extended by Laplace to the so-called central limit theorem (CLT), which is one of the most important results in probability
- Carl Friedrich Gauss in 1809 used the normal distribution to solve the important statistical problem of combining observations
- Almost all basic statistical inference is based on the normal probability distribution

3.2.4 Normal probability distribution

A random variable X is said to have a normal probability distribution with parameters μ and σ^2, if it has a pdf given by:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, x \in R, \mu \in R, \sigma > 0$$

- If $\mu = 0$, and $\sigma = 1$, we call it a standard normal random variable
- Theorem 3.2.5 Mean, Variance, and Moment-Generating Function of a Normal Random Variable

$$X \sim N(\mu, \sigma^2), E(X) = \mu, Var(X) = \sigma^2, M_X(t) = e^{t\mu + \frac{1}{2}t^2\sigma^2}$$

- z-transform (or z-score) of X

$$X \sim N(\mu, \sigma^2), Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$$

Example 3.2.8

- For $X \sim N(0,1)$, calculate $P(X > 1.13)$.
- For $X \sim N(5,4)$, calculate $P(2.5 < X < 10)$.

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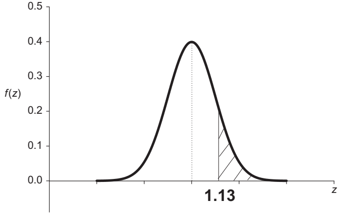
- For $X \sim N(5,4)$, calculate $P(2.5 < X < 10)$.

Solution

$$P(Z > 1.13) = 1 - P(Z \leq 1.13) = 1 - 0.8708 = 0.1292$$

R-code: `pnorm(1.13, mean=0, sd=1, lower.tail=FALSE)`



3.2.4 Normal probability distribution

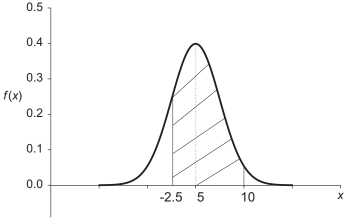
Using the z-transform, we have:

$$P(-2.5 < X < 10) = P\left(\frac{-2.5-5}{2} < \frac{X-5}{2} < \frac{10-5}{2}\right)$$

$$= P(-3.75 < Z < 2.5) = P(-3.75 < Z < 0) + P(0 < Z < 2.5) = 0.9938$$

R-code: pnorm(2.5, mean=0, sd=1, lower.tail=TRUE) - pnorm(-3.75, mean=0, sd=1, lower.tail=TRUE)

pnorm(10, mean=5, sd=2, lower.tail=TRUE) - pnorm(2.5, mean=5, sd=2, lower.tail=TRUE)



3.2.4 Normal probability distribution

Example 3.2.9 For a Standard Normal Random Variable Z, Find the value of z_0 such that:

(a) $P(Z > z_0) = 0.25$

(b) $P(Z < z_0) = 0.95$

(c) $P(Z < z_0) = 0.12$

(d) $P(Z > z_0) = 0.68$

Example 3.2.9 For a Standard Normal Random Variable Z, Find the value of z_0 such that:

(a) $P(Z > z_0) = 0.25$

(b) $P(Z < z_0) = 0.95$

(c) $P(Z < z_0) = 0.12$

(d) $P(Z > z_0) = 0.68$

Solution

(a) From the normal table, and using the fact that the shaded area in the figure is 0.25, we have:

$$P(Z > z_0) = 0.5 - P(0 \leq Z \leq z_0) = 0.25$$

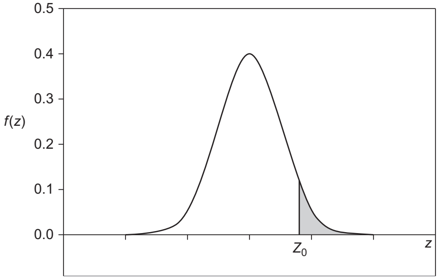
Thus, $P(0 \leq Z \leq z_0) = 0.25$, and hence, we obtain $z_0 \approx 0.675$.

Solution continued

(b) $P(Z < z_0) = 0.5 + P(0 \leq Z < z_0) = 0.95, P(0 \leq Z < z_0) = 0, 45.$ From the normal table: $z_0 = 1.645.$

(c) $P(0 < X < -z_0) = 0.38.$ From the normal table, we find $z_0 = -1.175.$

(d) $P(Z > z_0) = 0.5 + P(z_0 < Z < 0) = 0.68,$ $P(0 < Z < -z_0) = 0.18.$ From the normal table: $z_0 = -0.465.$



3.2.4 Normal probability distribution

Example 3.2.10 The scores of an examination are assumed to be normally distributed with $\mu = 75$ and $\sigma^2 = 64$. What is the probability that a student score chosen at random will be greater than 85?

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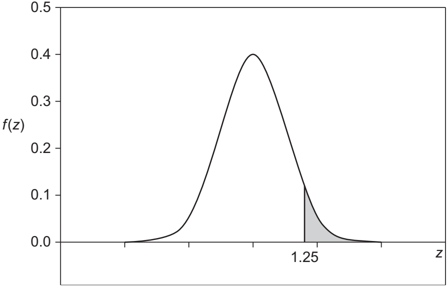
Solution

Let X be a randomly chosen score from the exam scores, $X \sim N(75, 64)$:

$$P(X > 85) = P\left(\frac{X - 75}{8} > \frac{85 - 75}{8}\right) = P\left(Z > \frac{1.25}{1}\right) = 0.1056$$



3.2.4 Normal probability distribution

When do we know that our data follow the normal distribution?

- If the histogram of our data can be capped with a bell-shaped curve
- If the stem-and-leaf diagram is fairly symmetrical with respect to its center
- quantile or QQ plot

QQ plot:scatterplot

- x axis: the quantiles of the scores
- y axis: the expected normal scores q_i by taking the z-score of $(r_i - 0.5)/n$. $(q_i = qnorm((r_i - 0.5)/n, 0, 1)$ in R, and $icdf('normal', (r_i - 0.5)/n, 0, 1)$ in matlab
  - r_i: the rank of the i_{th} observation in increasing order
  - n: the number of samples

3.2.4 Normal probability distribution

Steps for QQ plot

- Sort the data $(x_1, \cdots, x_n)$ as $(x_{(1)}, \cdots, x_{(n)})$ and calculate $r_i, q_i, i = 1, 2, \cdots, n$
- Plot of these scores $x_{(i)}$ against the expected normal scores q_i
- Plotted points fits a straight line: normal
- Plotted points donot fit the line well but bend away from it in places: nonnormal

3.2.4 Normal probability distribution

- normal probability plot

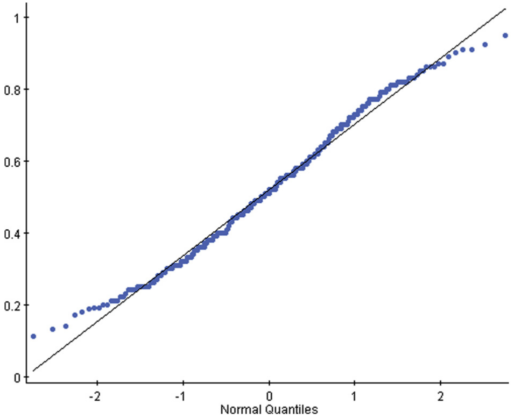


FIGURE 3.3 Normal probability plot.

3.2.4 Normal probability distribution

- normal probability plot

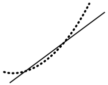
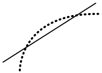
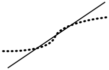
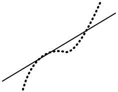
	If the layout of points appears to bend up and to the left of the normal line that indicates a long tail to the right, or right skew.
	If the layout of points bends down and to the right of the normal line that indicates a long tail to the left, or left skew.
	An S-shaped layout of points indicates shorter than normal tails, thus, a smaller variance is expected.
	If the layout of points starts below the normal line, bends to follow it, and ends above it indicates long tails. That is, there is more variance than we would expect in a normal distribution.

FIGURE 3.4 Shapes indicating distribution behavior of the data.

Log-normal probability distribution

- The pdf of a log-normal random variable, X, is given as:

$$f(x) = \begin{cases} \frac{1}{x\sigma_y\sqrt{2\pi}} e^{-\frac{(\ln x - \mu_y)^2}{2\sigma_y^2}}, x > 0, \\ \quad\quad\quad 0, \text{ otherwise} \end{cases} \quad \mu_y \in R, \sigma_y > 0$$

- If X is log-normally distributed with parameters μ_y and σ_y, then

$$Y = \ln(X) \sim N(\mu_y, \sigma_y^2)$$

  - μ_y and σ_y are mean and standard deviation of $Y = \ln(X)$

3.2.4 Normal probability distribution

- The expected value

$$E(X) = e^{\mu_y + (\sigma_y^2/2)}$$

- The variance

$$Var(X) = \left(e^{\sigma_y^2} - 1\right) e^{2\mu_y + \sigma_y^2}$$

- Get μ_y and σ_y from $E(X)$ and $Var(X)$

$$\mu_y = \ln\left(\sqrt{\frac{\mu_x^4}{\mu_x^2 + \sigma_x^2}}\right), \quad \sigma_y = \ln\left(\sqrt{\frac{\mu_x^2 + \sigma_x^2}{\mu_x^2}}\right)$$

3.2.4 Normal probability distribution

Example 3.2.11 In an effort to establish a suitable height for the controls of a moving vehicle, information was gathered about X, the amounts by which the heights of the operators vary from 60 in., which is the minimum height. It was verified that the data that were collected followed the log-normal distribution by a normal probability plot of $Y = \ln(X)$. Assume that $\mu_X = 6$ in. and $\sigma_X = 2$ in.

(a) What percentage of operators would have a height less than 65.5 in.?

(b) If an operator is chosen at random, what is the probability that his or her height will be between 64 and 66 in.?

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(a) What percentage of operators would have a height less than 65.5 in.?

(b) If an operator is chosen at random, what is the probability that his or her height will be between 64 and 66 in.?

Solution

- (a) Here, $X = 65.5 - 60 = 5.5$. Also,

$$\mu_Y = \ln\left(\sqrt{\frac{\mu_X^4}{\mu_X^2 + \sigma_X^2}}\right) = \ln\left(\sqrt{\frac{64^2}{64^2 + 2^2}}\right) = 1.74;$$

3.2.4 Normal probability distribution

$$\sigma_Y = \ln\left(\frac{\mu_X^2 + \sigma_X^2}{\mu_X^2\sqrt{s_X^2}}\right) = \ln\left(\frac{64^2 + 2^2}{64^2\sqrt{2^2}}\right) = 0.053.$$

Thus,

$$P(X \leq 5.5) = P(Y \leq \ln(5.5)) = P\left(\frac{Y - \ln(5.5)}{0.053} \leq \frac{1.74 - \ln(5.5)}{0.053}\right)$$
$$= P(Z \leq 0.67) = 0.2514.$$

Hence, about 25.14% of the heights of the operators vary from 60 in.
(b) Similar to (a), we get:

$$P(4 \leq X \leq 6) = P(\ln 4 \leq Y \leq \ln 6)$$

$$= P\left(\frac{\ln 4 - 1.74}{0.053} \leq Z \leq \frac{\ln 6 - 1.74}{0.053}\right) = P(6.67 \leq Z \leq 0.98) = 0.8365.$$

Thus, 83.65% of the heights of the operators will be between 64 and 66 in.

3.2.5 Gamma probability distribution

A random variable X is said to possess a gamma probability distribution with parameters $\alpha > 0$ and $\beta > 0$, if it has the pdf given by:

$$f(x) = \begin{cases} \dfrac{1}{\beta^{\alpha}\Gamma(\alpha)} x^{\alpha-1} e^{-\frac{x}{\beta}}, \; x > 0 \\ 0, \text{ otherwise} \end{cases}$$

with

$$\Gamma(\alpha) = \int_0^{\infty} e^{-x} x^{\alpha-1} dx, \alpha > 0, \Gamma(n) = (n-1)!$$

- $\Gamma(\alpha)$: Gamma function
- Denoted by $\Gamma(\alpha, \beta)$
    - α: shape parameter
    - β: scale parameter

3.2.5 Gamma probability distribution

- $\Gamma(\alpha, \beta)$

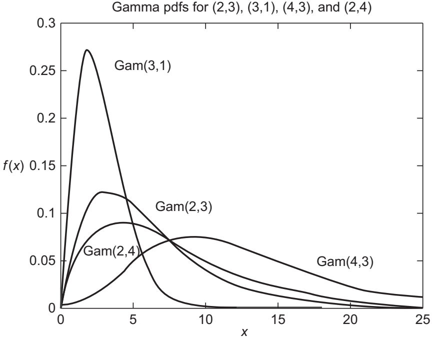

FIGURE 3.5 Gamma pdfs for different degrees of freedom.

- Mean, Variance, and Moment-Generating Function of a Gamma Random Variable

$$E(X) = \alpha\beta, Var(X) = \alpha\beta^2, M_X(t) = \frac{1}{(1-\beta t)^{\alpha}}, t < \frac{1}{\beta}$$

Example 3.2.12 The daily consumption of aviation fuel in millions of gallons at a certain airport can be treated as a gamma random variable with parameters $\alpha = 3$ and $\beta = 1$.

(a) What is the probability that on a given day the fuel consumption will be less than 1 million gallons?

(b) Suppose the airport can store only 2 million gallons of fuel. What is the probability that the fuel supply will be inadequate on a given day?

3.2.5 Gamma probability distribution

Solution

(a) Let X be the fuel consumption in millions of gallons on a given day at a certain airport. Then, $X \sim \Gamma(\alpha = 3, \beta = 1)$, and the probability density function is given by

$$f(x) = \frac{1}{\Gamma(3) \cdot (1^3)} x^{3-1} e^{-x} = \frac{1}{2} x^2 e^{-x}, \quad x > 0$$

Hence, using integration by parts, we obtain:

$$P(X < 1) = \frac{1}{2} \int_0^1 x^2 e^{-x} dx = 1 - \frac{5}{2} e = 0.08025$$

(b) Because the airport can store only 2 million gallons, the fuel supply will be inadequate if the fuel consumption X is greater than 2. Thus,

$$P(X > 2) = \frac{1}{2}\int_{2}^{\infty} x^2 e^{-x} dx = 0.677$$

3.2.5 Gamma probability distribution

Special Case: Exponential random variable A random variable X is said to have an exponential probability distribution with parameter β if the pdf of X is given by:

$$f(x) = \begin{cases} \frac{1}{\beta}e^{-\frac{x}{\beta}}, \ \beta > 0, 0 \leq x < \infty \\ 0, \text{ otherwise} \end{cases}$$

- $X \sim Exp(\beta)$
- If $\alpha = 1$, $X \sim \Gamma(\alpha, \beta)$ becomes an exponential probability distribution with parameter β.
- Mean, Variance, and Moment-Generating Function of an Exponential Random Variable

$$E(X) = \beta, Var(X) = \beta^2, M_X(t) = \frac{1}{(1-\beta t)}, t < \frac{1}{\beta}$$

3.2.5 Gamma probability distribution

Example 3.2.13 The time, in hours, during which an electrical generator is operational is a random variable that follows the exponential distribution with parameter $\beta = 160$. What is the probability that a generator of this type will be operational for:

(a) less than $40h$

(b) between 60 and $160h$

(c) more than $200h$

Example 3.2.13 The time, in hours, during which an electrical generator is operational is a random variable that follows the exponential distribution with parameter $\beta = 160$. What is the probability that a generator of this type will be operational for:

(a) less than $40h$

(b) between 60 and $160h$

(c) more than $200h$

Solution

(a) Let X denote the random variable corresponding to the time (in hours) during which the generator is operational. Then the density function of X is given by:

$$f(x) = \begin{cases} \dfrac{1}{160} e^{-\dfrac{x}{160}}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

Thus, we have the following:

$$P(X < 40) = \int_0^{40} \frac{1}{160} e^{-\frac{x}{160}} dx \approx 0.22119.$$

(b) Using the density function of X, we can calculate:

$$P(60 \leq X \leq 160) = \int_{60}^{160} \frac{1}{160} e^{-\frac{x}{160}} dx \approx 0.3194.$$

(c) To find the probability that X exceeds 200 hours, we calculate:

$$P(X > 200) = \int_{200}^{\infty} \frac{1}{160} e^{-\frac{x}{160}} dx \approx 0.2865.$$

Special case: Chi-square χ^2random variable

In the pdf of the gamma, $\alpha = \frac{n}{2}$ and $\beta = 2$, we have chi-square random variable $X \sim \chi^2(n)$

- The pdf of $\chi^2(n)$:

$$f(x) = \begin{cases} \frac{1}{\Gamma(\frac{n}{2})2^{\frac{n}{2}}} x^{\frac{n}{2}-1} e^{-\frac{x}{2}}, \ 0 \leq x < \infty \\ 0, \ \text{otherwise} \end{cases}$$

- Mean, Variance, and Moment-Generating Function of a Chi-Square Random Variable

$$E(X) = n, Var(X) = 2n, M_X(t) = \frac{1}{(1-2t)^{\frac{n}{2}}}, t < \frac{1}{2}$$

beta random variable

- A random variable X is said to have a beta distribution with parameters α and β if and only if the density function of X is given by:

$$f(x) = \begin{cases} \dfrac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}, 0 \le x \le 1, \alpha, \beta > 0 \\ 0, \text{ otherwise} \end{cases}$$

  - beta function: $B(\alpha,\beta) = \int_0^1 x^{\alpha-1}(1-x)^{\beta-1}dx = \dfrac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$

- Mean and Variance

$$E(X) = \frac{\alpha}{\alpha+\beta}, Var(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

Contents

3.3 Joint probability distributions

Definition 3.3.1 bivariate distribution

- Let X and Y be random variables. If both X and Y are discrete, then:

$$p(x, y) = P(X = x, Y = y)$$

  is called the joint probability function of X and Y

- If both X and Y are continuous, then $f(x, y)$ is called the joint probability density function (joint pdf) of X and Y if and only if:

$$P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f(x, y)dydx$$

Theorem 3.3.1 If X and Y are two random variables

- For discrete case

$$p(x,y) \geq 0, \sum_{x,y} p(x,y) = 1$$

- For continous case

$$f(x,y) \geq 0, \int\int f(x,y)dxdy = 1$$

Definition 3.3.2 marginal pdf (pmf)

- If X and Y are discrete, then the marginal pdfs (pmf) of X and Y, respectively, are denoted by

$$p_X(x) = \sum_{all\ \ y} P(X = x, Y = y), p_Y(y) = \sum_{all\ \ x} P(X = x, Y = y)$$

- If X and Y are continuous, then the marginal pdfs (pmf) of X and Y, respectively, are denoted by

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy, f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx,$$

Example 3.3.2

Find the marginal pdf of the random variables X and Y, if their joint probability function is given by Table 3.1.

TABLE 3.1 Joint pmf of *X* and *Y*.

	y				
x	−2	0	1	4	Sum
−1	0.2	0.1	0.0	0.2	0.5
3	0.1	0.2	0.1	0.0	0.4
5	0.1	0.0	0.0	0.0	0.1
Sum	0.4	0.3	0.1	0.2	1.0

Find the marginal densities of X and Y.

Example 3.3.2

Find the marginal pdf of the random variables X and Y, if their joint probability function is given by Table 3.1.

TABLE 3.1 Joint pmf of *X* and *Y*.

	y				
x	−2	0	1	4	Sum
−1	0.2	0.1	0.0	0.2	0.5
3	0.1	0.2	0.1	0.0	0.4
5	0.1	0.0	0.0	0.0	0.1
Sum	0.4	0.3	0.1	0.2	1.0

Find the marginal densities of X and Y.

Solution

x_i	−1	3	5	otherwise	y_i	−2	0	1	4	otherwise
$f_X(x_i)$	0.5	0.4	0.1	0	$f_Y(y_i)$	0.4	0.3	0.1	0.2	0

Definition 3.3.3 The conditional probability distribution of the random variable X given Y is given by:

- if X and Y are discrete

$$p(x|y) = p(x|Y = y) = \frac{p(x, y)}{p_Y(y)} = \frac{P(X = x, Y = y)}{p_Y(y)}$$

- if X and Y are continuous and $f_Y(y) \neq 0$

$$f(x|y) = f(x|Y = y) = \frac{f(x, y)}{f_Y(y)}$$

Definition 3.3.4 (independence of X and Y) Let X and Y have a joint pmf $p(x, y)$ or pdf $f(x, y)$. Then X and Y are independent if and only if:

- if X and Y are discrete

$$p(x, y) = p_X(x)p_Y(y)$$

- if X and Y are continuous

$$f(x, y) = f_X(x)f_Y(y)$$

Example 3.3.3. Let $f(x,y)$ be defined as follows:

$$f(x,y)=\begin{cases}3x, & \text{if } 0<y<x<1\\ 0, & \text{otherwise}\end{cases}$$

(a) Find $P\left(X\leq\frac{1}{2},\frac{1}{4}<Y<\frac{3}{4}\right)$.

(b) Find the marginal probability density functions $f_X(x)$ and $f_Y(y)$.

(c) Find the conditional probability density function $f(x|y)$ for $0<y<1$. Also compute $f(x|Y=\frac{1}{2})$.

(d) Are X and Y independent?

Solution

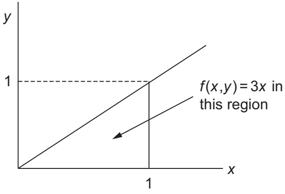

FIGURE 3.8 Domain of $f(x, y)$.

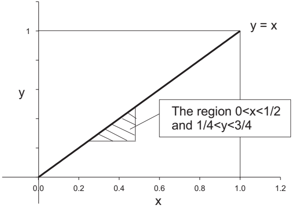

FIGURE 3.9 Region of integration.

(a)

$$P\left(X \leq \frac{1}{2}, \frac{1}{4} < Y < \frac{3}{4}\right) = \int_{1/4}^{1/2}\int_{1/4}^{x} 3x dy dx = \int_{1/4}^{1/2} 3x\left(x - \frac{1}{4}\right)dx$$

$$= \left(\frac{3x^3}{3} - \frac{3x^2}{8}\right)\Bigg|_{1/4}^{1/2} = \frac{5}{128}.$$

(b) To find the marginals, we note that for each x, y varies from 0 to x $(0 < y < x)$. Therefore,

$$f_X(x) = \int_0^x 3x\,dy = 3x^2, \quad 0 < x < 1.$$

Similarly, for each y, x varies from y to 1.

$$f_Y(y) = \int_y^1 3x\,dx = \frac{3}{2}(1 - y^2), \quad 0 < y < 1.$$

(c) Using the definition of conditional density:

$$f(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{3x}{\frac{3}{2}(1-y^2)} = \frac{2x}{1-y^2}, \quad y \le x \le 1.$$

Thus, from this we have:

$$f(x|y = 1/2) = \frac{8}{3}x, \quad \frac{1}{2} \le x \le 1.$$

(d) To check for independence of X and Y:

$$f_X(1) \cdot f_Y\left(\frac{1}{2}\right) = (3) \cdot \left(\frac{9}{8}\right) = \frac{27}{8} \neq f(1, 1/2).$$

Hence, X and Y are not independent.

Definition 3.3.5 The expected value of $g(X, Y)$

- if X and Y are discrete

$$E\left[g(X,Y)\right] = \sum_{x,y} g(x,y)p(x,y)$$

- if X and Y are continuous

$$E\left[g(X,Y)\right] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y)f(x,y)dxdy$$

Properties of Expected Value

- $E(aX + bY) = aE(X) + bE(Y)$
- If X and Y are independent, then $E(XY) = E(X)E(Y)$. However, the converse is not necessarily true.

Example 3.3.4

Let $f(x, y)$ be defined as follows:

$$f(x,y) = \begin{cases} 3x, & \text{if } 0 \leq y \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find $E(4X - 3Y)$.

(b) Find $E(XY)$.

Example 3.3.4

Let $f(x, y)$ be defined as follows:

$$f(x,y) = \begin{cases} 3x, & \text{if } 0 \leq y \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find $E(4X - 3Y)$.

(b) Find $E(XY)$.

Solution

(a) $E(X) = \int_0^1 x \cdot 3x^2 \, dx = \frac{3}{4}$,

$E(Y) = \int_0^1 y \cdot \frac{3}{2}(1 - y^2) \, dy = \frac{3}{8}$.

Hence, $E(4X - 3Y) = 4 \cdot \frac{3}{4} - 3 \cdot \frac{3}{8} = \frac{15}{8}$.

(b) $E(XY) = \int_0^1 \int_0^x xy \cdot 3x \, dy \, dx = \frac{3}{10}$.

Definition 3.3.6 the conditional expectation of $g(X)$ given $Y = y$

- if X and Y are discrete

$$E\left[g(X)|y\right] = E\left[g(X)|Y=y\right] = \sum_{all\ \ x} g(x)p(x|y)$$

- if X and Y are continuous

$$E\left[g(X)|y\right] = E\left[g(X)|Y=y\right] = \int_{-\infty}^{\infty} g(x)f(x|y)dx$$

-

$$Var(X|y) = E\left[(X - E(X|y))^2\,|y\right] = E\left[X^2|y\right] - \left[E(X|y)\right]^2$$

Some remarks about the conditional expectation of $g(X)$ given $Y = y$

- $E\left[g(X)|y)\right]$ is a function of y
- If we let Y range over all of its possible values, the conditional expectation $E\left[g(X)|Y\right]$ is a function of the random variable Y

Theorem 3.3.2

Let X and Y be two random variables. Then:

$$E(X) = E\left[E(X|Y)\right]$$

$$Var(X) = E\left[Var(X|Y)\right] + Var\left[E(X|Y)\right]$$

Example 3.3.5 Let X and Y be two random variables with a joint density function given by:

$$f(x,y) = \begin{cases} (x^2 + \frac{1}{3}xy), & \text{if } 0 \leq x \leq 1 \text{ and } 0 \leq y \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

Find the conditional expectation, $E(X|Y = \frac{1}{2})$.

Example 3.3.5 Let X and Y be two random variables with a joint density function given by:

$$f(x,y) = \begin{cases} (x^2 + \frac{1}{3}xy), & \text{if } 0 \le x \le 1 \text{ and } 0 \le y \le 2 \\ 0, & \text{otherwise} \end{cases}$$

Find the conditional expectation, $E(X|Y = \frac{1}{2})$.

Solution

First, we will find the conditional density, $f(x|y)$. The marginal PDF is given by:

$$f_Y(y) = \int_0^1 \left(x^2 + \frac{xy}{3}\right) dx = \frac{1}{3} + \frac{1}{6}y, \quad 0 < y < 2.$$

Therefore,

$$f(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{x^2 + xy}{\frac{1}{6}y + \frac{1}{3}}, \quad 0 \le x \le 1.$$

Hence,

$$f_{X|Y}(x|y = \frac{1}{2}) = \frac{x^2 + x}{\frac{1}{12} + \frac{1}{3}} = \frac{12}{5}\left(x^2 + \frac{x}{6}\right).$$

Thus,

$$E(X|Y = \frac{1}{2}) = \int_0^1 x\left(\frac{12}{5}x^2 + \frac{x}{6}\right)\, dx = \frac{11}{15} \approx 0.733.$$

Example 3.3.6

Let the joint density of two random variables X and Y be given by:

$$f(x,y) = \begin{cases} \frac{1}{4}(2x+y), & \text{if } 0 \le x \le 1 \text{ and } 0 \le y \le 2 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find $f_X(x)$ and $f_Y(y)$.

(b) Find Var(X).

(c) Find $E(X|Y)$ and Var$(X|Y)$.

Example 3.3.6

Let the joint density of two random variables X and Y be given by:

$$f(x,y) = \begin{cases} \frac{1}{4}(2x+y), & \text{if } 0 \le x \le 1 \text{ and } 0 \le y \le 2 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find $f_X(x)$ and $f_Y(y)$.

(b) Find $\text{Var}(X)$.

(c) Find $E(X|Y)$ and $\text{Var}(X|Y)$.

Solution

(a) We have:

$$f_X(x) = \int_0^2 \frac{1}{4}(2x+y)\,dy = \frac{1}{4}(4x+2), \quad 0 \le x \le 1.$$

Similarly, the marginal PDF of Y, $f_Y(y)$ is:

$$f_Y(y) = \int_0^1 \frac{1}{4}(2x+y)\,dx = \frac{1}{4}(1+y), \quad 0 \le y \le 2.$$

(b) To find the variance,

$$E(X) = \int_0^1 \frac{1}{4}x(4x+2)\,dx = \frac{7}{12},$$

and

$$E(X^2) = \int_0^1 \frac{1}{4}x^2(4x+2)\,dx = \frac{5}{12}.$$

Thus, the variance of X is:

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = \frac{5}{12} - \left(\frac{7}{12}\right)^2 = \frac{11}{144}.$$

(c) First, find the conditional density of X given that $Y = y$,

$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{2x+y}{1+y}, \quad 0 \le x \le 1; 0 \le y \le 2.$$

Then, the conditional expectation is given by:

$$E[X|Y=y] = \int_0^1 x\frac{2x+y}{1+y}\,dx = \frac{1}{1+y}\int_0^1 (2x^2+xy)\,dx = (\frac{1}{6})(\frac{4+3y}{1+y})$$

For the conditional variance, we also need to find:

$$E[X^2|Y=y] = \int_0^1 x^2 \cdot \frac{2x+y}{1+y}\,dx = \frac{1}{1+y}\int_0^1 (2x^3+x^2y)\,dx$$
$$= \frac{1}{6} \cdot \frac{3+2y}{1+y}.$$

Thus,

$$\text{Var}[X|Y=y] = E[X^2|Y=y] - [E(X|Y=y)]^2 = \frac{3y^2+6y+2}{36(1+y)^2}.$$

3.3.1 Covariance and correlation

Definition 3.3.7

- The covariance between two random variables X and Y:

$$\sigma_{XY} = Cov(X, Y) = E\left[(X - \mu_X)(Y - \mu_Y)\right] = E(XY) - \mu_X \mu_Y$$

  where $\mu_X = E(X)$ and $\mu_Y = E(Y)$

- The correlation coefficient ρ_{XY}:

$$\rho_{XY} = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$$

  - Correlation is the measure of the linear relationship between the random variables X and Y. If $Y = aX + b, (a \neq 0)$,then $|\rho_{XY}| = 1$

Properties of Covariance and Correlation Coefficient

- $-1 \leq \rho_{XY} \leq 1$
- If X and Y are independent, then $\rho_{XY} = 0$.
  - The converse is not true
- If $Y = aX + b$, then:

$$\rho_{XY} = \begin{cases} 1, \text{ if } a > 0 \\ -1, \text{ if } a < 0 \end{cases}$$

3.3.1 Covariance and correlation

Properties of Covariance and Correlation Coefficient

- If $U = a_1 X + b_1$ and $V = a_2 Y + b_2$, then

$$Cov(U, V) = a_1 a_2 Cov(X, Y)$$

- - $Var(aX + bY) = a^2 Var(X) + b^2 Var(Y) + 2abCov(X, Y)$
  - If X and Y are independent

$$Var(aX + bY) = a^2 Var(X) + b^2 Var(Y)$$

- If $X_1, \ldots, X_n$ are independent, then

$$Var(\sum_{i=1}^{n} X_i) = \sum_{i=1}^{n} Var(X_i)$$

Example 3.3.7 The joint probability density function (PDF) of the random variables X and Y is given by:

$$f(x,y) = \begin{cases} \frac{1}{64} e^{-\frac{y}{8}}, & \text{if } 0 \leq x \leq y \leq \infty \\ 0, & \text{otherwise} \end{cases}$$

To find the covariance of X and Y, we will use the formula:

$$\text{Cov}(X,Y) = E(XY) - E(X)E(Y)$$

where $E(X)$ and $E(Y)$ are the expected values (means) of X and Y, respectively, and $E(XY)$ is the expected value of their product.

Solution

$$E(XY) = \int_0^\infty \int_0^y xy \cdot \frac{1}{64} e^{-\frac{y}{8}} \, dx \, dy = \frac{1}{64} \int_0^\infty y e^{-\frac{y}{8}} \left(\int_0^y x \, dx \right) \, dy$$

$$= \frac{1}{128} \int_0^\infty y^3 e^{-\frac{y}{8}} \, dy = 192.$$

$$E(X) = \int_0^\infty \int_0^y x \cdot \frac{1}{64} e^{-\frac{y}{8}} \, dx \, dy = 8,$$

$$E(Y) = \int_0^\infty \int_0^y y \cdot \frac{1}{64} e^{-\frac{y}{8}} \, dx \, dy = 16.$$

Thus, $\mathrm{Cov}(X, Y) = 192 - (8)(16) = 64.$

3.3.1 Covariance and correlation

Definition 3.3.8 Let X and Y be jointly distributed. Then the joint moment-generating function is defined by:

$$
\begin{aligned}
&M_{(\boldsymbol{X},\boldsymbol{Y})}(t_{\mathbf{1}}, t_{\mathbf{2}}) \\
&= E\left(e^{t_1 X + t_2 Y}\right) \\
&= \begin{cases} \sum\limits_y \sum\limits_x e^{t_1 x + t_2 y} p(x,y), & \text{if } X \text{ and } Y \text{are discrete} \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{t_1 x + t_2 y} f(x,y) dx dy, & \text{if } X \text{ and } Y \text{are continuous} \end{cases}
\end{aligned}
$$

Contents

The velocity V of a gas molecule (Maxwell-Boltzmann law) behaves as a gamma-distributed random variable. We would like to derive the distribution of $E = mV^2$, the kinetic energy of the gas molecule

- If We are given the distribution of X, how can we find the distribution of $Y = g(X)$?

3.4.1 Method of distribution functions

Procedure to Find the Cumulative Distribution Function of a Function of a Random Variable Using the Method of Distribution Functions

- if $Y = g(X_1, \dots, X_n)$, then we can summarize the method of distribution function as follows
  - Find the region$\{Y \leq y\}$ in the $(x_1, x_2, \dots, x_n)$ space, that is, find the set of $(x_1, x_2, \dots, x_n)$ for which $g(x_1, x_2, \dots, x_n) \leq y$
  - Find $F_Y(y) = P(Y \leq y)$ by integrating joint pdf $f(x_1, x_2, \dots, x_n)$ over the region $\{Y \leq y\}$
  - Find the density function $f_Y(y)$ by differentiating $F_Y(y)$

3.4.1 Method of distribution functions

Example 3.4.1

Let $X \sim \mathcal{N}(0,1)$. Using the cdf of X, find the pdf of X^2.

Example 3.4.1

Let $X \sim \mathcal{N}(0,1)$. Using the cdf of X, find the pdf of X^2.

Solution

Let $Y = X^2$. Note that the marginal PDF of X is given by:

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \quad x \in (-\infty, \infty)$$

Then, the CDF of Y for a given $y \geq 0$ is:

$$F(y) = P(Y \leq y) = P(X^2 \leq y) = P\left(-\sqrt{y} \leq X \leq \sqrt{y}\right)$$

$$= \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\, dx = 2\int_0^{\sqrt{y}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\, dx, \text{(by the symmetry of } e^{-x^2/2}\text{)}.$$

3.4.1 Method of distribution functions

Hence, by differentiating $F(y)$ with respect to y, we obtain the marginal pdf of Y:

$$f_{\mathbf{Y}}(y) = \begin{cases} \frac{2}{\sqrt{2\pi}} e^{-\frac{y}{2}} \frac{1}{2\sqrt{y}} & \text{for } 0 < y < \infty, \\ 0 & \text{otherwise.} \end{cases}$$

This is the PDF of a $\chi^2(1)$ with 1 degree of freedom.

Example 3.4.2

Suppose that the random variable X has a Poisson probability distribution, that is,

$$f(x) = \begin{cases} \frac{e^{-\lambda} \cdot \lambda^x}{x!} & \text{for } x = 0, 1, 2, \dots, \\ 0 & \text{otherwise.} \end{cases}$$

Find the cumulative distribution function (cdf) of $Y = aX + b$.

Solution :

The cdf of Y is given by:

$$F(y) = P(Y \leq y) = P(aX + b \leq y) = P\left(X \leq \frac{y-b}{a}\right) = \sum_{x=0}^{\left[\frac{y-b}{a}\right]} \frac{e^{-\lambda} \cdot \lambda^x}{x!},$$

where $[x]$ is the largest integer less than or equal to x. Therefore

$$F(y) = \begin{cases} 0 & \text{if } y < b, \\ \sum_{x=0}^{\left[\frac{y-b}{a}\right]} \frac{e^{-\lambda} \cdot \lambda^x}{x!} & \text{if } y \geq b. \end{cases}$$

It should be noted that the pmf $f_Y(y)$, can be found from the equation:

$$f_Y(y) = F_Y(y) - F_Y(y-1), for\ y = an + b, where\ n = 0, 1, 2, \cdots$$

3.4.2 Methods of function

The probability density function of $Y = g(X)$, where g is differentiable and monotone increasing or decreasing

- $Y = g(X)$ where X is a continuous random variable with pdf $f_X(x)$
- g is differentiable and the inverse function g^{-1} of g exists and
- $X = g^{-1}(Y)$
- The density function of Y can be given by

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

3.4.2 Methods of function

Example 3.4.3 Let $X \sim N(0,1)$. Find the probability density function (pdf) of $Y = e^X$.

Example 3.4.3 Let $X \sim N(0,1)$. Find the probability density function (pdf) of $Y = e^X$.

Solution

Here $g(x) = e^x$, and hence, $g^{-1}(y) = \ln(y)$. Thus, $\frac{d}{dy} g^{-1}(y) = \frac{1}{y}$.

Also, $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$, $-\infty < x < \infty$.

Therefore, the pdf of Y is:

$$f_Y(y) = \begin{cases} \frac{1}{y\sqrt{2\pi}} e^{-\frac{\ln^2(y)}{2}}, & \text{for } y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

3.4.3 Probability integral transformation

- Let X be a continuous random variable, with pdf $f_X(x)$ and cdf $F(x)$
- Let $Y = F(X)$
- then

$$P(Y \leq y) = P\left(F(X) \leq y\right) = P\left(X \leq F^{-1}(y)\right)$$

$$= \int_{-\infty}^{F^{-1}(y)} f_X(x)dx = F_X(x)|_{-\infty}^{F^{-1}(y)} = y$$

- hence

$$f(y) = \begin{cases} 1, \ 0 < y < 1 \\ 0, \ \text{otherwise} \end{cases}$$

3.4.3 Probability integral transformation

Example 3.4.4

- Let X be a normal random variable with mean m and variance σ^2. Thus, the probability density function of X is given by:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty, \quad -\infty < \mu < \infty, \text{and} \quad \sigma^2 > 0.$$

Let $Y = \int_{-\infty}^{X} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(u-\mu)^2}{2\sigma^2}} du$. Then $Y = F(X)$, where F is the cdf of a standard normal random variable. Therefore, Y follows a uniform on $(0, 1)$. That is:

$$f(y) = \begin{cases} 1, & \text{if } 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

3.4.4 Functions of several random variables

Method of distribution functions

- Let $X_1, \ldots, X_n$ be continuous independent and identically distributed (iid) random variables with pdf $f(x)$ (cdf $F(x)$)
- The pdf of $Y_1 = \min(X_1, \ldots, X_n)$

$$F_{Y_1}(y) = 1 - (1 - F(y))^n, f_{Y_1}(y) = n(1 - F(y))^{n-1} f(y)$$

- The pdf of $Y_n = \max(X_1, \ldots, X_n)$

$$F_{Y_n}(y) = (F(y))^n, f_{Y_n}(y) = n(F(y))^{n-1} f(y)$$

3.4.5 Transformation method

We illustrate the method for bivariate distributions

- Let the joint pdf of (X, Y) be $f(x, y)$. Let $U = g_1(X, Y)$ and $V = g_2(X, Y)$
- The mapping from (X, Y) to (U, V) is assumed to be one to one and onto. Hence, there are functions, h_1 and h_2, such that:

$$x = h_1^{-1}(u, v), y = h_2^{-1}(u, v)$$

- Define the Jacobian of the transformation J by:

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

- Then the joint pdf of U and V is given by:

$$f(u, v) = f\left(h_1^{-1}(u, v), h_2^{-1}(u, v)\right)|J|$$

Example 3.4.6 Let X and Y be independent random variables with a common probability density function (pdf) $f(x) = e^{-x}$ for $x > 0$. Find the joint pdf of $U = \frac{X}{X+Y}$ and $V = X + Y$.

Solution

We have $U = \frac{X}{X+Y} = \frac{X}{V}$. Hence, $X = UV$ and $Y = V - X = V(1 - U)$. Thus, the Jacobian is given by:

$$J = \begin{vmatrix} v & u \\ -v & (1-u) \end{vmatrix}$$

Then $|J| = v(1-u) + uv = v(> 0)$. Note that $0 \leq u \leq 1$, $0 < v < \infty$, and the joint pdf of U and V is:

$$f(u, v) = f\left(h_1^{-1}(u, v), h_2^{-1}(u, v)\right) \cdot |J| = e^{-uv} \cdot e^{-v(1-u)} \cdot v$$

$$= ve^{-v}, \quad 0 \leq u \leq 1, \quad 0 < v < \infty,$$

Example 3.4.7

Let X and Y be independent random variables uniformly distributed on $[0,1]$. Find the pdf of $X+Y$.

Solution

$$\begin{aligned} U &= X+Y, \\ V &= Y, \\ f(x,y) &= 1, 0 \le x \le 1, 0 \le y \le 1 \\ X &= U-V \\ Y &= V \end{aligned}$$

$$J = \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1$$

Thus, we have the joint pdf (U,V),

$$f(u,v) = \begin{cases} 1 & \text{if } 0 \le u-v \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

Solution continued

Note that because$V$ is the variable we introduced, to get the pdf of U, we just need to find the marginal pdf from the joint pdf. the regions of integration are $0 \leq u \leq 1$ and $0 \leq u \leq 2$

$$f_U(u) = \int f(u,v)dv = \begin{cases} \int_0^u dv = u, & 0 \leq u \leq 1, \\ \int_{u-1}^1 dv = 2-u, & 0 \leq u \leq 2. \end{cases}$$

Contents

3.5 Limit theorems

Chebyshev's Theorem Let the random variable X have a mean μ and standard deviation σ. Then for $K > 0$, a constant,

$$P\{|X-\mu| < K\sigma\} \geq 1 - \frac{1}{K^2}$$

or

$$P\{|X-\mu| \geq K\sigma\} \leq \frac{1}{K^2}$$

- A random variable X differs from its mean by at least K standard deviations is less than or equal to $\frac{1}{K^2}$
- We can also write Chebyshev's theorem as:

$$P\{|X-\mu| \geq \epsilon\} \leq \frac{Var(X)}{\epsilon^2} = \frac{\sigma^2}{\epsilon^2}$$

for some $\epsilon > 0$

Example 3.5.1 A random variable X has mean 24 and variance 9. Obtain a bound on the probability that the random variable X assumes values between 16.5 and 31.5.

Solution

From Chebyshev's theorem:

$$P(\mu - K\sigma < X < \mu + K\sigma) \geq 1 - \frac{1}{K^2}$$

Equating $\mu + K\sigma$ to 31.5 and $\mu - K\sigma$ to 16.5 with $\mu = 24$ and $\sigma = \sqrt{9} = 3$, we obtain $K = 2.5$. Hence,

$$P(16.5 < X < 31.5) \geq 1 - \frac{1}{(2.5)^2} = 0.84$$

Example 3.5.2 Let X be a random variable that represents the systolic blood pressure of the population of 18- to 74-year-old men in the United States. Suppose that X has mean 129 mm Hg and standard deviation 19.8 mm Hg.

(a) Obtain a bound on the probability that the systolic blood pressure of this population will assume values between 89.4 and 168.6 mm Hg.

(b) In addition, assume that the distribution of X is approximately normal. Using the normal table, find $P(89.4 \leq X \leq 168.6)$. Compare this with the empirical rule.

Solution

(a) Because we are given only the mean and standard deviation, and no probability distribution is specified, we can use Chebyshev's theorem. We have:

$$P(\mu - K\sigma < X < \mu + K\sigma) \geq 1 - \frac{1}{K^2}$$

Equating $\mu + K\sigma$ to 168.6 and $\mu - K\sigma$ to 89.4 with $\mu = 129$ and $\sigma = 19.8$, we obtain $K = 2$. Hence,

$$P(89.4 \leq X \leq 168.6) \geq 1 - \frac{1}{2^2} = 0.75$$

(b) Because X is normally distributed with mean 129 and standard deviation 19.8, using the z-score, we get:

$$P(89.4 \leq X \leq 168.6) = P\left(\frac{89.4 - 129}{19.8} \leq Z \leq \frac{168.6 - 129}{19.8}\right)$$

$$= P(-2 \leq Z \leq 2) = 0.9544$$

Hence, approximately 95.44% of this population will have systolic blood pressure values between 89.4 and 168.6 mm Hg. This compares well with the 95% value from the empirical rule.

Law of Large Numbers

Theorem 3.5.2 Let $X_1, \dots, X_n$ be a set of pairwise independent random variables with $E(X_i) = \mu$, and $Var(X_i) = \sigma^2$. Then for any $c > 0$

$$P\left\{\mu - c \le \bar{X} \le \mu + c\right\} \ge 1 - \frac{\sigma^2}{nc^2}$$

and as $n \to \infty$, the probability approaches 1.
Equivalently,

$$P\left(\left|\frac{S_n}{n} - \mu\right| < \epsilon\right) \to 1$$

as $n \to \infty$. Here $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ and $S_n = \sum_{i=1}^{n} X_i$

Example 3.5.3 Let $X_1, X_2, \ldots, X_n$ be iid (i.i.d.) Bernoulli random variables with parameter p. Verify the weak law of large numbers.

Solution

For Bernoulli random variables, we know that $E(X_i) = p$ and $Var(X_i) = p(1-p)$. Thus, by Chebyshev's theorem, we have:

$$P\left\{(p - c \le \frac{S_n}{n} \le p + c)\right\} = P\left\{\left|\frac{S_n}{n} - p\right| \le c\right\} \ge 1 - \frac{\sigma^2}{nc^2}$$

$$= 1 - \frac{p(1-p)}{nc^2} \to 1 \quad \text{as } n \to \infty$$

This verifies the Weak Law of Large Numbers.

Example 3.5.4

Consider n rolls of a balanced die. Let X_i be the outcome of the ith roll, and let $S_n = \sum_{i=1}^{n} X_i$. Show that, for any $\epsilon > 0$,

$$P\left(\left|\frac{S_n}{n} - \frac{7}{2}\right| \geq \epsilon\right) \to 0 \quad \text{as } n \to \infty.$$

Solution

Because the die is balanced, $E(X_i) = \frac{7}{2}$. By the law of large numbers, for any $\epsilon > 0$,

$$P\left(\left|\frac{S_n}{n} - \frac{7}{2}\right| \geq \epsilon\right) \to 0 \quad \text{as } n \to \infty,$$

or equivalently,

$$P\left(\left|\frac{S_n}{n} - \frac{7}{2}\right| < \epsilon\right) \to 1 \quad \text{as } n \to \infty.$$

Central Limit Theorem

Theorem 3.5.3

If $X_1, \ldots, X_n$ is a random sample from an infinite population with mean $\mu < \infty$, and variance $\sigma^2 < \infty$, then the limiting distribution of $Z_n = \dfrac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$ as $n \to \infty$ is the standard normal probability distribution. That is,

$$\lim_{n \to \infty} P(Z_n \leq z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-\frac{t^2}{2}} dt$$

- $Z_n = \dfrac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} = \dfrac{S_n - n\mu}{\sigma \sqrt{n}}$

- Example 3.5.5

  Let $X_1, X_2, \ldots$ be iid random variables such that:

$$X_i = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1-p \end{cases}$$

  Show that $Z_n = \frac{S_n - np}{\sqrt{npq}}$ is approximately normal for large n, where $S_n = \sum_{i=1}^{n} X_i$, and $q = 1 - p$.

- Solution

  We know that:

$$E(X) = p; \quad E(X^2) = p; \quad Var(X) = p(1-p) = pq$$

  Hence, by the CLT, the limiting distribution of $Z_n = \frac{S_n - np}{\sqrt{npq}}$ as $n \to \infty$ is the standard normal probability distribution.

- Example 3.5.6
  A soft drink vending machine is set so that the amount of drink dispensed is a random variable with a mean of 8 oz and a standard deviation of 0.4 oz. What is the approximate probability that the average of 36 randomly chosen fills exceeds 8.1 oz?

- Solution
  From the CLT,
  $$\frac{(\bar{X}-8)}{\left(\frac{0.4}{\sqrt{36}}\right)} \sim \mathcal{N}(0,1).$$
  Hence, from the normal table,
  $$P\left(\bar{X}>8.1\right)=P\left(Z>\frac{8.1-8.0}{\frac{0.4}{\sqrt{36}}}\right)=P\left(Z>1.5\right)=0.0668.$$
  Thus, there is an approximately 6.68% chance that the chosen fills exceed 8.1 oz.

Example 3.5.7 Numbers in decimal form are often approximated by the closest integer. Suppose n numbers $X_1, \dots, X_n$ are approximated by their closest integers $J_1, J_2, \dots, J_n$. Let $U_i = X_i - J_i$. Assume that U_i are uniform on $(-0.5, 0.5)$ and that U_i are independent.

(a) Show that
$$\left(\frac{\sum_{i=1}^{n} U_i}{\sqrt{n/12}}\right) \sim \mathcal{N}(0,1) \text{ as } n \longrightarrow \infty$$

(b) For $n = 300$, find
$$P\left\{\frac{-5}{\sqrt{\frac{300}{12}}} \le \frac{\sum_{i=1}^{n} U_i}{\sqrt{\frac{300}{12}}} \le \frac{-5}{\sqrt{\frac{300}{12}}}\right\}$$

(c) For $n = 300$, find the value of a such that $P\{-a < \sum U_i < a\} = 0.95$.

(d) For $n = 10^6$, find a such that $P\{-a < \sum_{i=1}^{10^6} U_i < a\} = 0.99$.

Solution

(a) Because U_i's are uniform in $(-0.5, 0.5)$, $\sum U_i = 0$, and $\mathrm{Var}(U_i) = 1/12$. Let $S_n = \sum_{i=1}^{n} X_i$ and $K_n = \sum_{i=1}^{n} J_i$. Then:

$$P(|S_n - K_n| \leq a) = P(-a \leq \sum (X_i - J_i) \leq a) = P(-a \leq \sum U_i \leq a)$$

By the CLT,

$$\left(\frac{\sum_{i=1}^{n} U_i - 0}{\sqrt{n/12}} \right) \sim \mathcal{N}(0,1) \text{ as } n \longrightarrow \infty$$

(b) For $n = 300$ and $a = 5$, using the normal table,

$$P\left(\frac{-5}{\sqrt{\frac{300}{12}}} \leq \frac{\sum_{i=1}^{n} U_i}{\sqrt{\frac{300}{12}}} \leq \frac{5}{\sqrt{\frac{300}{12}}} \right) = 0.68.$$

(c) Now,

$$0.95 = P(-a \leq \sum U_i \leq a) = P\left(\frac{-a}{\sqrt{\frac{300}{12}}} \leq Z \leq \frac{a}{\sqrt{\frac{300}{12}}}\right).$$

From the normal table, we get $\frac{a}{\sqrt{\frac{300}{12}}} = 1.96$. This implies that $a = 9.8$.

(d) We have

$$0.99 = P\left(-a \leq \sum_{i=1}^{10^6} U_i \leq a\right) = P\left(\frac{-a}{\sqrt{\frac{10^6}{12}}} \leq Z \leq \frac{a}{\sqrt{\frac{10^6}{12}}}\right).$$

Now, using the normal table, we have $\frac{a}{\sqrt{\frac{10^6}{12}}} = 2.58$. Hence, $a = 745$.

Example 3.5.8 A casino has a coin, suspected to be biased. Estimate p (probability of heads) such that they can be 99% confident that their estimate (say, $\hat{p}$) is within 0.01 of p (unknown). What is the minimum number of times we need to toss this coin?

Solution

Set

$$X_j = \begin{cases} 1, & \text{if H appears in the } j\text{'th toss} \\ 0, & \text{if T appears in the } j\text{'th toss} \end{cases}$$

Suppose we decided to use $\hat{p} = \frac{\sum X_i}{n}$, that is, $\hat{p} = \frac{\#\text{Heads}}{n}$ as an estimate of p.

We want $P\left(|\bar{X} - p| < 0.01\right) = 0.99$.

Because $Y = \sum_{i=1}^{n} X_i \sim \text{Bin}(n, p)$, we have $E(Y) = np$, $Var(Y) = npq$.

By the CLT, $\frac{(\bar{X} - p)}{\sqrt{\frac{pq}{n}}} \sim N(0, 1)$

Now,

$$0.99 = P\left\{\frac{-0.01}{\sqrt{\frac{pq}{n}}} \le \frac{|\bar{X}-p|}{\sqrt{\frac{pq}{n}}} < \frac{0.01}{\sqrt{\frac{pq}{n}}}\right\} = P\left\{\frac{-0.01}{\sqrt{\frac{pq}{n}}} \le Z < \frac{0.01}{\sqrt{\frac{pq}{n}}}\right\}$$

Using the normal table, we find that $\frac{0.01}{\sqrt{\frac{pq}{n}}} = 2.58$, which implies that

$$\sqrt{n} \ge \left(\frac{2.58\sqrt{pq}}{0.01}\right)$$

Because the maximum of pq is $1/4$, it is sufficient that

$$\sqrt{n} = \frac{2.58\sqrt{1/4}}{0.01} = 129$$

Hence, $n \ge (129)^2 = 16,641$, and we should choose the sample size $n \ge 16,641$ Thus, the coin must be tossed at least 16,641 times to determine if the coin is not fair.

Contents

3.6 Chapter summary

- Some special probability distribution functions that arise in practice
  - Binomial
  - Poisson
  - Uniform
  - Normal (or Gaussian)
  - Exponential
- The behavior of joint probability distributions for more than one random variable
- How to find the probability (mass) density function and cumulative distribution for the functions of a random variable
- Chebyshev inequality, the law of large numbers, and the CLT for the random variables

Contents

3.7 Computer examples (optional)

- 3.7.1 The R examples
- 3.7.2 Minitab examples
- 3.7.3 Distribution checking
- 3.7.4 SPSS examples
- 3.7.5 SAS examples

Contents

Projects for Chapter 3

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- 3C Coupon collector's problem
- 3D Recursive calculation of binomial and Poisson probabilities
- 3E Simulation of Poisson approximation of binomial
- 3F Generating a large amount of random data using R

3A Mixture distribution

- Mixture distributions are used frequently in medical applications, such as microarray analysis
- Suppose a random variable X has pdf $f_1(x)$ with probability p_1 and pdf $f_2(x)$ with probability p_2, where $p_1 + p_2 = 1$. Then we say that the random variable X has a mixture distribution
- This can be thought of as observing a Bernoulli random variable Z that is equal to 1 with probability p_1 and 2 with probability p_2

$$X = \begin{cases} X_1 \sim f_1(x),\ Z = 1 \\ X_2 \sim f_2(x),\ Z = 2 \end{cases}$$

Projects for Chapter 3

- This can be thought of as observing a Bernoulli random variable Z that is equal to 1 with probability p_1 and 2 with probability p_2

$$X = \begin{cases} X_1 \sim f_1(x),\ Z = 1 \\ X_2 \sim f_2(x),\ Z = 2 \end{cases}$$

- the pdf of X is given by $f(x) = p_1 f_1(x) + p_2 f_2(x)$
- if μ_1, σ_1^2 and μ_2, σ_2^2 are means and variances of $f_1(x)$ and $f_2(x)$, respectively, then

$$\mu = E(X) = p_1\mu_1 + p_2\mu_2$$

$$\sigma^2 = Var(X) = p_1\sigma_1^2 + p_2\sigma_2^2 + p_1\mu_1^2 + p_2\mu_2^2 - (p_1\mu_1 + p_2\mu_2)^2$$

Projects for Chapter 3

- Proof.

$$F(x) = P(X \le x) = P(X_1 \le x)P(Z = 1) + P(X_2 \le x)P(Z = 2)$$

$$= p_1 F_1(x) + p_2 F_2(x)$$

then

$$f(x) = p_1 f_1(x) + p_2 f_2(x)$$

- Proof.

$$E(X) = E\left[E(X|Z)\right] = E(X|Z = 1)p_1 + E(X|Z = 2)p_2$$

$$= E(X_1)p_1 + E(X_2)p_2 = p_1\mu_1 + p_2\mu_2$$

$$Var(X) = \cdots \text{ this is for you}$$

3B Generating samples from exponential and Poisson probability distribution

- Generate a sample from $\frac{1}{\theta}e^{-\frac{x}{\theta}}$ (θ is chosen). Let $Y_1, \cdots, Y_n$ be a sample from a $U(0,1)$ distribution.
  Let $F(x) = 1 - e^{-\frac{x}{\theta}}$ (cdf of exponential). $Y = F(X)$ is uniform.
- $y_j = 1 - e^{-\frac{x_j}{\theta}}$ implies $x_j = -\theta \ln(1 - y_j) = -\theta \ln u_j$ where $u_1, u_2, \cdots, u_n$ is a sample from $U(0,1)$. $x_1, \cdots, x_n$ is a sample from an exponential distribution with parameter θ
- Suppose we want to generate a sample from a Poisson probability distribution with parameter λ. $X_1, \cdots, X_n$ is a sample from an exponential distribution with parameter $\frac{1}{\lambda}$ until $\sum_{i=1}^{n} X_i$ just exceeds 1. Then $y_n = (n-1)$ is a sample value from a Poisson probability distribution with parameter λ ???

Projects for Chapter 3

3C Coupon collector's problem

- Suppose there are n distinct colors of coupons. Each coupon color is equally likely to occur. When a complete set of coupons with each color represented is assembled, you win a prize. Let $X =$ number coupons for a complete set.
- Find (a) distribution of X, (b) $E(X)$, and (c) $Var(X)$.

3D Recursive calculation of binomial and Poisson probabilities

- A simple way to calculate binomial probabilities is as follows: For a given n and p, evaluate $b(0, n, p)$ and then apply the recursive relationship:

$$b(x+1, n, p) = b(x, n, p)\frac{p(n-x)}{(1-p)(x+1)}$$

  to obtain other binomial probabilities

- The following recursive formulas are very useful in calculating successive Poisson probabilities:

$$f(x-1, \lambda) = f(x, \lambda)\frac{x}{\lambda}$$

$$f(x+1, \lambda) = f(x, \lambda)\frac{\lambda}{x+1}$$

Projects for Chapter 3

3E Simulation of Poisson approximation of binomial

- Write and run R-code (or others) with various n and p to see how the errors compare as n increases and p decreases, by calculation of actual binomial probabilities as well as Poisson probabilities with $\lambda = np$

3F Generating a large amount of random data using R

- For each of the generated random samples, write an R-code (or others) to create a histogram overlapped by the corresponding density function

Exercises and Projects

- Exercises
  - Exercises 3.2.5, 3.2.15, 3.2.18, 3.2.21
  - Exercises 3.3.6, 3.3.8, 3.3.9
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  - Exercises 3.5.4, 3.5.6, 3.5.13
- Projects
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